

TRI-COUNTY AP, NC, USA

WMO: 723079

Lat: 36.298N

Lon: 77.171W

Elev: 21

StdP: 101.08

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 93796

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.5	-4.9	-18.9	0.7	-0.3	-15.7	1.0	-0.1	7.7	8.9	6.5	7.5	0.9	350	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	34.9	24.8	32.9	24.7	32.2	24.4	26.8	31.9	26.0	31.1	25.3	30.3	1.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	20.5	29.5	24.5	19.5	28.7	23.8	18.7	27.8	83.7	32.0	80.5	31.0	77.2	30.4	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.3	4.6	DB	-11.7	35.3	3.8	2.3	-14.5	37.0	-16.7	38.3	-18.8	39.5	-21.6	41.2	(4)
(5)			WB	-12.4	27.9	3.5	1.1	-14.9	28.7	-16.9	29.3	-18.8	29.9	-21.3	30.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	5.0	6.7	10.1	15.4	19.7	24.0	25.9	25.3	22.1	16.2	10.5	7.0	(6)	
	DBStd	8.65	6.26	5.15	5.40	4.82	3.98	3.20	2.62	2.73	3.31	4.78	5.05	5.44	(7)	
	HDD10.0	560	180	117	68	8	0	0	0	0	0	6	54	126	(8)	
	HDD18.3	1846	413	326	259	113	33	2	0	0	9	99	240	352	(9)	
	CDD10.0	2647	26	26	72	171	301	420	492	476	364	198	68	34	(10)	
	CDD18.3	891	1	1	4	26	76	172	233	217	123	33	4	2	(11)	
	CDH23.3	7682	4	7	44	251	658	1549	2204	1873	834	226	23	7	(12)	
	CDH26.7	3007	0	0	4	64	213	641	973	792	264	54	1	1	(13)	
(14) Wind	WSAvg	1.5	1.5	1.6	1.9	1.9	1.5	1.3	1.1	1.2	1.4	1.3	1.4	1.3	(14)	
(15) Precipitation	PrecAvg	1189	87	70	100	83	93	126	139	121	127	80	72	88	(15)	
	PrecMax	1563	165	146	189	164	166	274	252	278	570	163	190	174	(16)	
	PrecMin	897	28	13	23	34	41	64	62	33	12	0	8	36	(17)	
	PrecStd	174	38	35	39	33	34	61	49	64	119	45	46	34	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.7	24.1	27.3	31.3	32.6	36.0	36.3	36.4	33.7	31.3	26.1	24.0	(19)	
		MCWB	18.2	17.0	17.6	21.0	23.8	24.1	24.5	25.7	23.2	23.1	19.0	19.9	(20)	
	2%	DB	21.3	21.8	24.1	28.5	31.1	33.7	35.0	34.2	31.3	27.8	23.1	21.8	(21)	
		MCWB	17.0	16.4	16.8	19.3	22.4	24.0	25.1	24.8	23.1	20.9	17.9	17.8	(22)	
	5%	DB	17.9	19.0	22.4	26.2	29.0	32.2	32.8	32.4	29.5	26.0	22.1	18.9	(23)	
		MCWB	14.4	14.6	15.3	18.0	21.4	23.9	25.2	24.9	23.2	19.9	17.4	16.4	(24)	
	10%	DB	14.9	16.5	20.0	23.9	27.2	30.8	32.1	31.2	27.9	23.8	19.5	16.4	(25)	
		MCWB	11.4	11.6	14.8	17.5	20.5	23.8	24.8	24.6	22.1	19.8	16.0	13.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.4	19.2	20.7	23.3	25.1	26.8	27.8	28.2	25.9	24.9	21.5	20.8	(27)	
		MCDB	22.1	22.5	23.5	28.4	30.9	32.3	32.7	33.6	29.6	28.4	23.4	23.5	(28)	
	2%	WB	17.7	17.5	18.7	21.3	23.6	25.7	26.8	26.9	24.9	23.3	19.5	18.7	(29)	
		MCDB	20.5	20.4	22.3	25.2	28.4	31.5	31.9	31.9	28.6	26.1	22.1	20.9	(30)	
	5%	WB	15.2	15.6	17.2	19.9	22.7	24.9	26.0	25.9	24.2	22.0	18.0	16.7	(31)	
		MCDB	17.8	18.8	20.6	24.0	27.0	30.3	31.3	30.6	27.6	24.4	20.5	18.4	(32)	
	10%	WB	12.2	12.4	15.5	18.7	21.6	24.1	25.3	25.1	23.3	20.5	16.4	14.0	(33)	
		MCDB	14.0	15.4	19.2	22.8	25.7	28.9	30.6	29.5	26.4	23.0	19.5	16.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	11.7	12.3	12.8	11.5	10.7	10.0	9.9	10.1	11.8	12.4	10.9	(35)	
		MCDBR	13.6	15.2	15.0	14.5	13.2	12.2	11.5	11.5	11.8	12.5	13.4	13.0	(36)	
	5% WB	MCWBR	10.2	10.6	8.8	7.2	6.4	5.5	5.1	5.2	5.7	6.7	8.7	10.1	(37)	
		MCWBR	12.7	13.9	12.5	12.4	11.5	10.7	10.5	10.1	9.5	9.7	11.0	11.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.304	0.319	0.349	0.390	0.438	0.492	0.519	0.513	0.406	0.357	0.322	0.307	(40)	
		taud	2.527	2.481	2.429	2.322	2.218	2.115	2.054	2.059	2.356	2.479	2.530	2.553	(41)	
	Ebn at Noon		898	921	919	893	851	802	776	772	848	866	867	871	(42)	
		Edh at Noon	79	93	107	126	142	158	166	162	115	93	79	73	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.47	3.19	4.19	5.34	5.95	6.36	6.04	5.46	4.47	3.60	2.74	2.18	(44)	
	RadStd		0.19	0.36	0.32	0.40	0.46	0.39	0.30	0.31	0.36	0.45	0.22	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (64 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+157	+106

Nomenclature: See separate page